

Brown leaf spot of rice

Cochliobolus miyabeanus (*Bipolaris oryzae*)

	Prevention	Monitoring	Direct Control	Direct Control	Restrictions
 Oval spots with brown/light brown grey centres (Donald Groth, Louisiana State University AgCenter, Bugwood.org)  Brown lesions on rachis (Chin Khoon Min)  Milled rice grains showing symptoms of infection (Donald Groth, Louisiana State University AgCenter, Bugwood.org)	• Use good quality certified seed • Transplant only seedlings that are healthy (free from brown spot symptoms) • Destroy volunteer rice crop and infected rice debris by burning soon after harvest • Control weeds in season and out of season • Amend soil with calcium silicate to boost resistance to the disease • Seed treatment: pre-soak seeds in cold water for 8 hours and treat with hot water (53–54°C) for 10–12 minutes before planting OR treat with fungicide for prevention before sowing (see yellow control column for details)	• Monitor crop weekly through the season as disease can set in at all stages of growth. On the leaves and leaf sheaths look for small circular lesions, dark brown to purple in colour. Older lesions are more oval with a light brown to grey center surrounded by a reddish brown margin • Check glumes and panicle rachis for dark brown to black oval spots or discoloration on the surface • Brown spot lesions can be mistaken for blast lesions. Lesions caused by brown spot are oval while blast lesions are diamond or spindle shape • As soon as symptoms appear start control measures. Fungicides can also be applied as a preventative measure before the disease is seen	• Rogue infected plants from the nursery to avoid disease spread and burn to destroy • Rogue plants with brown spot symptoms at low disease incidence in the field after transplanting	• When using a pesticide or botanical, always wear protective clothing and follow the instructions on the product label • Do not use chemicals with the same mode of action year after year as this can lead to resistance • Always consult the most recent list of registered pesticides of MOFA, Ghana • Apply Mancozeb (e.g. Agrithane 80WP, Benco 80WP, Damazeb 80WP, Dizcozeb 80WP etc) at a rate of 75g/15L. Multi-site broad spectrum contact fungicide. FRAC group: M3 • Apply Fosetyl-AI (e.g. Folpan) at 35–40g/15L. Systemic fungicide, FRAC group: 33 • Apply Carbendazim (e.g. Carbendazim 50WP, Goldazim 500SC) at a rate of 25ml/15L. Broad spectrum systemic fungicide, FRAC group: B1 • Seed treatment: Folpet (Folpan 50 WP) at 3–4g/kg of seed. FRAC code: M4	• WHO Class U (Unlikely to present acute hazard in normal use). When symptoms persist or there are signs of spreading apply 3 times in a season at 7–14 day intervals. REI-24hrs; PHI-14 days. • WHO Class U (Unlikely to present acute hazard in normal use). When symptoms persist or there are signs of disease spreading Apply 3 times in a season at 7–10 day intervals. REI-24hrs; PHI-14 days. • WHO Class U (Unlikely to present acute hazard in normal use). When symptoms persist and there are signs of disease spreading Apply 2 times at 14 day intervals. REI-24hrs; PHI-14 days. • WHO Class U (Unlikely to present acute hazard in normal use)



Ghana

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